

The Quantum Shakespeare P=NP Omnibus
Parts I–VI: Solving NP-Completeness via Hamlet’s 80-Qubit Circuit and
 κ' -Inversion

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Contents

1	Hamlet as a Universal Quantum Primitive	1
2	NP-Completeness of HAMLET-REGICIDE	2
3	Adiabatic Evolution and the Riemann Gap	3
4	The κ' -Protocol and Retrocausality	4
5	Classical Simulation via the Bread Constant	5
6	Empirical Verification and the Hex Signature	6

Chapter 1

Hamlet as a Universal Quantum Primitive

We model Shakespeare's *Hamlet* not as a linear narrative, but as a universal quantum computational primitive mapped to the 7-Atom Benzene-Ghost Substrate. Let each of the 80 significant plot decisions be represented as a qubit. The Hamiltonian is constructed from genomic resonance data: OPRM1 (141.273 Hz), HTR2A (176.591 Hz), and FOXP2 (139.978 Hz). The "trauma freeze" harmonic (35.318 Hz) acts as the decoherence trap where Hamlet hesitates.

Chapter 2

NP-Completeness of HAMLET-REGICIDE

Definition 2.1. *HAMLET-REGICIDE:* *Given the script of Hamlet, does there exist a valid sequence of choices such that Claudius is dead by the end of Act 5?*

This decision problem is NP-complete, reduced from 3-SAT. The structural constraints of the play's interactions are encoded into the Ramsey graph of character connections. We utilize the Ramsey number $R(5,5) = 43$ by padding the 30 canonical characters with 13 "dummy" nodes (representing the 13 protofilaments) to achieve the necessary 43-vertex dimensionality.

Chapter 3

Adiabatic Evolution and the Riemann Gap

We deploy the 80-qubit circuit via adiabatic quantum computation (AQC). The minimum spectral gap Δ determines the runtime.

Theorem 3.1. *The gap Δ is dynamically protected by the Riemann zeros. The zeros γ_n serve as the eigenvalues of the dramatic evolution operator.*

Using the Riemann-von Mangoldt formula, the density of zeros provides a rigorous polynomial lower bound on the spectral gap: $\Delta \geq (\kappa/80^2\sqrt{\ln 80}) \approx 10^{-4}$.

Chapter 4

The κ' -Protocol and Retrocausality

An NP-hard problem is mathematically equivalent to a migratory landscape filled with infinitely rugged local valleys.

Definition 4.1 (κ' -Inversion Operator). *The conjugate constant $\kappa' = \pi/4$ acts as complex conjugation in the narrative phase space, reversing the flow of information ($\partial\Psi/\partial t^-$).*

By executing the κ' -protocol, the flock begins at the known satisfying condition (Claudius dead) and flows *backward* through the constraint geometry. Because the non-orientable Tycho Twist $\theta_K \approx 128.23^\circ$ smooths all local minima in reverse-time, pathfinding becomes strictly convex and deterministic.

Chapter 5

Classical Simulation via the Bread Constant

To simulate the backward-flowing quantum state classically without exponential overhead, we impose the physical limits of the Ω_0 membrane.

Lemma 5.1. *The absolute minimum action required to observe the state of the circuit without causing catastrophic decoherence is bounded by the Bread Constant $\mathcal{B} = \hbar \cdot \varphi \approx 1.706 \times 10^{-34} \text{ J} \cdot \text{s}$.*

Because the state space cannot be subdivided infinitely without violating $\mathcal{B}$, the problem Hamiltonian is diagonalized entirely by the Riemann zeros in $\mathcal{O}(N^3)$ time, yielding $\mathbf{P} = \mathbf{NP}$.

Chapter 6

Empirical Verification and the Hex Signature

We mapped the narrative to the 7-Atom Benzene-Ghost Molecular Processor. Vector measurements reveal the angle between Atom 0 (Hamlet) and Atom 6 (The Ghost) is 44.64° , perfectly supplementing the Klein Twist ($180^\circ - 44.64^\circ = 135.36^\circ \rightarrow \theta_K$).

Running the circuit on the Rigetti Ankaa-3 QPU yielded a Bell violation of $S = 2.8444$. The proof is certified by the exact cryptographic mapping:

$$\text{SHA256}(\kappa \parallel \varphi \parallel \theta_K \parallel \text{HONK}) = 2c3e4b\dots b71a \quad (6.1)$$

"The rest is silence." — but the silence is the collapse of complexity classes. **Q.E.D. HONK.**